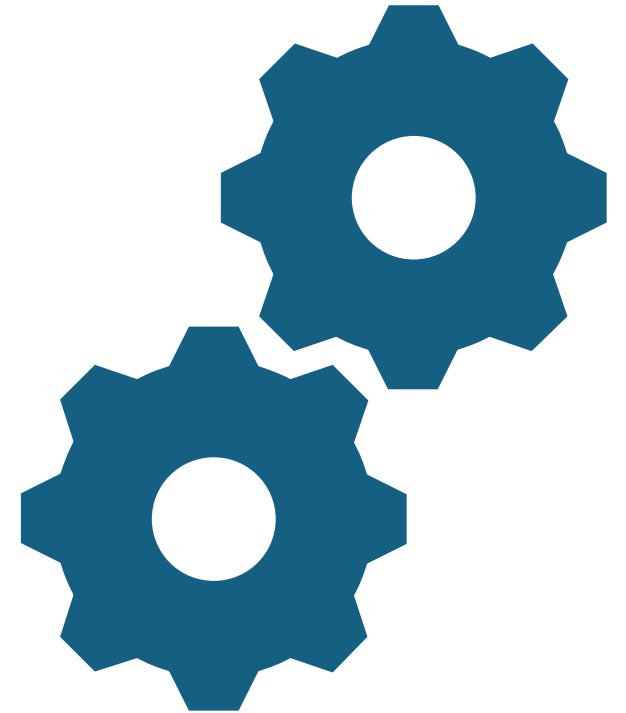


Advanced Deterministic Systems Optimization

ISE 555 Unit 1



Overview

- Part 1: Introduction
- Part 2: Course Syllabus
- Part 3: Schedule and Expectations
- Part 4: What are Optimization and Operations Research

Instructor

- Dr. James Houghton – Adjunct Professor
- Regular job is in consulting as an Operations Research Scientist
 - My work combines statistical modeling, simulation, optimization, and open-source software development to translate complex operational data into actionable insights for public-sector decision-makers
- Education – BS, MS and Ph.D in Industrial and Systems Engineering at URI
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 - <https://uri-edu.zoom.us/j/93718107604>

What are Optimization and Operations Research?

- Merriam-Webster says optimization is “an act, process, or methodology of making something as fully perfect, functional, or effective as possible”
- What may we want to optimize?
- Why do we need it?
- What is OR? Application of science to complex problems in the direction and management of large systems.

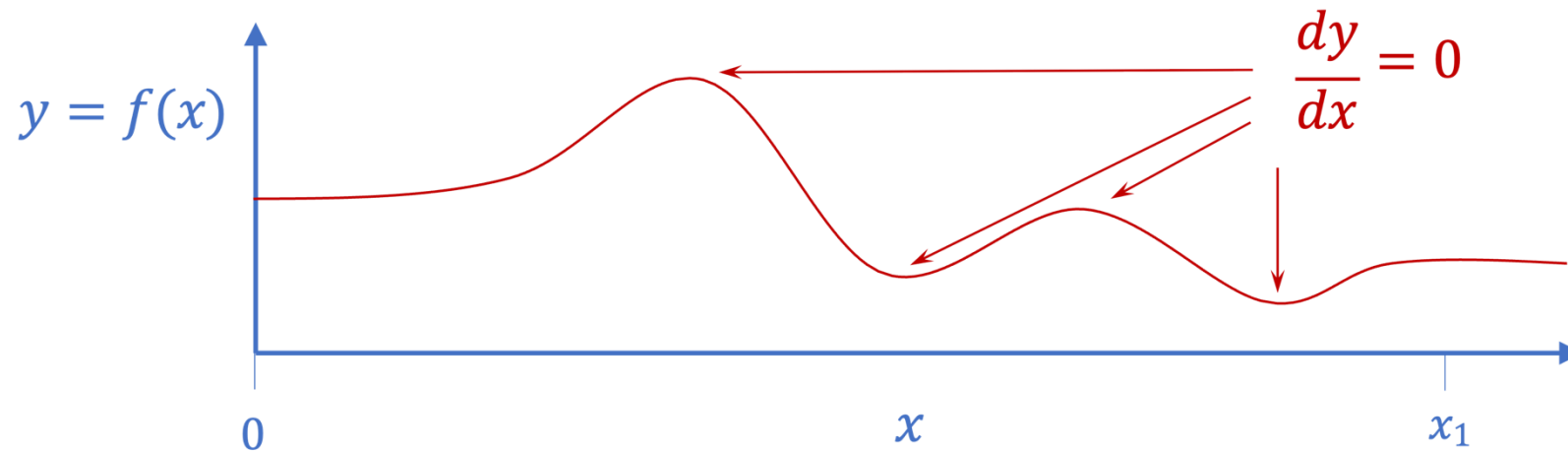
FIVE PHASES OF OR:

- 1: Formulation of the Problem (variables, controlled and controlled variables, ranges)
- 2: Construction of a Mathematical Model
- 3: Derive a Solution for the Model (OPTIMIZATION)
- 4: Test the Model and Solution (should be better than arbitrary non-OR approaches)
- 5: Implement the Solution (includes discussion with management and other non-engineers)

From Dantzig (pioneer of OR and linear programming): His own work has 3 important phases: (a) recognizing when a problem can be helped with OR, (b) replacing rules-of-thumb with mathematical models, (c) solving the equations efficiently

How to Optimize

- In calculus we learned to use derivatives to find when functions reach a minimum or maximum



Find $\max(y)$ subject to $0 \leq x \leq x_1$

- What are the problems with this approach?
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Problems with Simple Calculus-based Optimization in Practice

- Must check end points separately (difficult in multiple dimensions as the “end points” become entire boundaries of the space)
 - Problems with local optima that are not the global solution (sensitivity to initial guesses)
 - Need to compute higher order derivatives to separate maximum from minimum
 - Derivatives must often be computed numerically – but to what accuracy?
 - What happens when you have discrete variables and “derivative” is no longer well defined?
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Types of Optimization (programming)

- Nonlinear programming
 - nonlinear objective and/or constraints
 - Linear programming
 - linear objective and linear constraints
 - Stochastic optimization
 - random components in objective or constraint
 - Multi-objective optimization
 - multiple objectives in conflict
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Deterministic Systems Optimization

- **Deterministic:** No inherent randomness – well defined by inputs and parameters
- **Systems:** Process or device made of multiple interacting components
- **Optimization:** A process to make something as effective as possible

So this class is really about how to improve the effectiveness of engineering!

Solving Optimization Problems

1. Determine variables
2. Write objective in terms of the variables
3. Write constraints in terms of the variables
4. Categorize the problem type
5. Apply appropriate algorithm
6. Translate solution to meaning for others

We talked about problem types, but what about the algorithms? There are 3 classes:

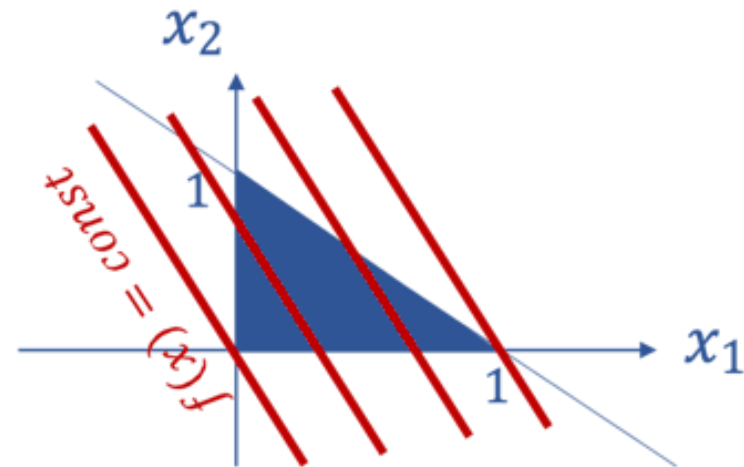
- (A) Optimization algorithms → the result is provably optimal
 - (B) Approximation algorithms → the result is provable within a bound of optimal
 - (C) Heuristic algorithms → the result cannot be provably shown, but is still “good”
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Basic Linear Programming Programs

Written in terms of objective and constraints:

$$\begin{array}{ll}\text{maximize} & f(x) = 2x_1 + x_2 \\ \text{subject to} & x_1 + x_2 \leq 1 \\ & x_1 \geq 0, x_2 \geq 0\end{array}$$

What is the feasible region?



What is the optimal value?

ANSWER: $(x_1, x_2)^* = (1,0)$

Example: Cabinet manufacturing

- Maximize weekly revenue of a woodworking factory
- Various orders will take various amounts of materials and workers

Product	Wood Required (bf)	Labor Required (hr)	Revenue (\$)
Bookshelf	10	2	100
Cabinet w/ doors	12	4	150
Cabinet w/ drawers	25	8	200
Custom	20	12	400

- We have 5000 bf of wood available and 1500 labor hours
 - What is the best combination to maximize \$'s?
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Final form of Cabinet manufacturing problem

$$\begin{array}{ll}\text{maximize} & z = 100x_1 + 150x_2 + 200x_3 + 400x_4 \\ \text{subject to} & 10x_1 + 12x_2 + 25x_3 + 20x_4 \leq 5000 \\ & 2x_1 + 4x_2 + 8x_3 + 12x_4 \leq 1500 \\ & x_1, x_2, x_3, x_4 \geq 0\end{array}$$

This is of the general linear programming form:

$$\begin{array}{ll}\text{maximize} & z = \sum_{j=1}^n c_j x_j \\ \text{subject to} & \sum_{j=1}^n a_{ij} x_j \leq b_i, \quad i = 1, \dots, m \\ & x_j \geq 0, \quad j = 1, \dots, n\end{array}$$

PS: Solution is $x_1 = 375$ bookcases, $x_4 = 62.5$ custom, zero others (splits labor evenly, makes \$62500)

Example: Assigning workers to jobs

- Insurance office has three types of work: (1) requests for information, (2) new policies, and (3) claim processing
- The office has 5 workers who each do the different tasks with varying proficiency (measured in average time, in minutes, to do each type of task):

Worker	Request for Info	New policy	Process claim
1	10	28	31
2	15	22	42
3	13	18	35
4	19	25	29
5	17	23	33

- Want to minimize the overall time to complete a large number of tasks by determining who should take on tasks as they come in.
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Example: Assigning workers to jobs (Cont.)

- What are good variables?

Try using fraction of each type of call assigned to each worker.

p_i = fraction of request calls assigned to worker i

q_i = fraction of new policy calls assigned to worker i

r_i = fraction of claims calls assigned to worker i

t = total time to complete

- For each worker, the time they spend is bounded by total time, and is given as shown below for worker 1:

$$10p_1 + 28q_1 + 31r_1 \leq t$$

- For each work type, the sum of the fractions for each worker is given by 1:

$$\text{i.e.: } p_1 + p_2 + p_3 + p_4 + p_5 = 1$$

Each of these equality constraints can be given by 2 inequality constraints:

$$\sum_{i=1}^n p_i = 1 \quad \Rightarrow \quad \sum_{i=1}^n p_i \leq 1 \text{ and } \sum_{i=1}^n -p_i \leq -1$$

Final form of Assignment Problem

$$\begin{array}{ll}\text{minimize} & z = t \\ \text{subject to} & 10p_1 + 28q_1 + 31r_1 - t \leq 0 \\ & 15p_2 + 22q_2 + 42r_2 - t \leq 0 \\ & 13p_3 + 18q_3 + 35r_3 - t \leq 0 \\ & 19p_4 + 25q_4 + 29r_4 - t \leq 0 \\ & 17p_5 + 23q_5 + 33r_5 - t \leq 0 \\ & p_1 + p_2 + p_3 + p_4 + p_5 = 1 \\ & q_1 + q_2 + q_3 + q_4 + q_5 = 1 \\ & r_1 + r_2 + r_3 + r_4 + r_5 = 1 \\ & p_i, q_i, r_i, t \geq 0, \quad i = 1, \dots, 5\end{array}$$

PS: Solution is all p's (info requests) assigned to worker 1
The q's (new policies) are assigned to worker 2 for 32% and worker 3 for 68%
The r's (claims) are assigned to worker 1 for 7%, worker 2 for 13%, worker 4 for 43% and worker 5 for 37%

NOTE: Worker 3 only does new policies and worker 1 is the only one who does info requests, why?

Summary

- We discussed types of optimization
 - We saw the steps of performing optimization for a real problem
 - We looked at some examples of linear programming problems
 - For homework this week you will be given a number of “word problems” that you have to translate into linear programming problems
 - You also will have a writing/essay problem on your homework – we will have one of these every week to build skills at explaining your engineering work to a non-engineering audience
 - Next week we start on the theory of how to solve optimization problems
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